

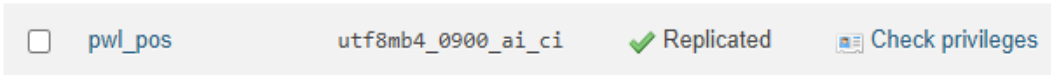
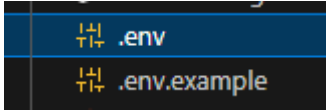


Mata Kuliah : Pemrograman Web Lanjut (PWL)  
Program Studi : D4 – Teknik Informatika  
Semester : 4

Kelas : TI  
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Jobsheet Ke- : 3

## Laporan Jobsheet

### Praktikum Ke-1 Pengaturan Database

| Langkah | Jawaban/Deskripsi                                                                                                                                                                                                                                                                                                    |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       |                                                                                                                                                                                                                                   |
| 3       |                                                                                                                                                                                                                                   |
| 5       | <pre>APP_NAME=Laravel APP_ENV=local APP_KEY=base64:alvxdU1906eEE0Vio2+2IjvNdudbVgAiK+qbF4mwMPE= APP_DEBUG=true APP_URL=http://localhost  LOG_CHANNEL=stack LOG_DEPRECATIONS_CHANNEL=null LOG_LEVEL=debug  DB_CONNECTION=mysql DB_HOST=127.0.0.1 DB_PORT=3306 DB_DATABASE=laravel DB_USERNAME=root DB_PASSWORD=</pre> |



## Praktikum Ke-2 Migration

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.1. | Pembuatan file migrasi tanpa relasi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 3    | <pre>λ php artisan make:migration create_m_level_table --create=m_level</pre> <p>INFO Migration [D:\laragon\www\WebLanjut\PWLP05\database\migrations\2025_02_24_053757_create_m_level_table.php] created successfully.</p>                                                                                                                                                                                                                                                                                                                                                                                         |
| 4    | <pre>public function up(): void {     Schema::create('m_level', function (Blueprint \$table) {         \$table-&gt;id('level_id');         \$table-&gt;string('level_kode', 10)-&gt;unique();         \$table-&gt;string('level_nama', 100);         \$table-&gt;timestamps();     }); }</pre>                                                                                                                                                                                                                                                                                                                     |
| 5    | <pre>2025_02_24_053757_create_m_level_table ..... 1,523ms DONE</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 8    | <pre>λ php artisan make:migration create_m_kategori_table</pre> <p>INFO Migration [D:\laragon\www\WebLanjut\minggu3\PWLP05\database\migrations\2025_02_25_092324_create_m_kategori_table.php] created successfully.</p> <pre>Schema::create('m_kategori', function (Blueprint \$table) {     \$table-&gt;id('kategori_id');     \$table-&gt;string('kategori_kode', 10)-&gt;unique();     \$table-&gt;string('kategori_nama', 100);     \$table-&gt;timestamps(); });</pre> <pre>λ php artisan migrate</pre> <p>INFO Running migrations.</p> <pre>2025_02_25_092324_create_m_kategori_table ..... 265ms DONE</pre> |
| 2.1. | Pembuatan file migrasi dengan relasi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |



2

```
return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('m_user', function (Blueprint $table) {
            $table->id('user_id');
            $table->unsignedBigInteger('level_id')->index();
            $table->string('username', 20)->unique();
            $table->string('nama', 100);
            $table->string('password');
            $table->timestamps();

            $table->foreign('level_id')->references('level_id')->on('m_
        ));
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::table('m_user', function (Blueprint $table) {
            schema::dropIfExists('m_user');
        });
    }
}
```

3

```
λ php artisan migrate --seed
INFO Running migrations.
2025_02_25_094825_create_m_user_table ..... 752ms DONE
```

4

```
D:\laragon\www\WebLanjut\minggu3\PWLP05(main -> origin)
λ php artisan make:migration create_m_barang_table
INFO Migration [D:\laragon\www\WebLanjut\minggu3\PWLP05\database\migrations\2025_02_25_154159_create_m_barang_table.php] created successfully.

D:\laragon\www\WebLanjut\minggu3\PWLP05(main -> origin)
λ php artisan make:migration create_t_penjualan_table
INFO Migration [D:\laragon\www\WebLanjut\minggu3\PWLP05\database\migrations\2025_02_25_154212_create_t_penjualan_table.php] created successfully.

D:\laragon\www\WebLanjut\minggu3\PWLP05(main -> origin)
λ php artisan make:migration create_t_stok_table
INFO Migration [D:\laragon\www\WebLanjut\minggu3\PWLP05\database\migrations\2025_02_25_154220_create_t_stok_table.php] created successfully.

D:\laragon\www\WebLanjut\minggu3\PWLP05(main -> origin)
λ php artisan make:migration create_t_penjualan_detail_table
INFO Migration [D:\laragon\www\WebLanjut\minggu3\PWLP05\database\migrations\2025_02_25_154228_create_t_penjualan_detail_table.php] created successfully.

λ php artisan migrate
INFO Running migrations.

2025_02_25_154159_create_m_barang_table ..... 508ms DONE
2025_02_25_154212_create_t_penjualan_table ..... 639ms DONE

2025_02_25_154220_create_t_stok_table ..... 785ms DONE
2025_02_25_154228_create_t_penjualan_detail_table ..... 729ms DONE
```



## Praktikum Ke-3 Seeder

| Langkah                                     | Jawaban/Deskripsi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |               |            |            |            |            |                                             |   |     |               |      |      |                                             |   |     |         |      |      |                                             |   |     |             |      |      |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------|------------|------------|------------|------------|---------------------------------------------|---|-----|---------------|------|------|---------------------------------------------|---|-----|---------|------|------|---------------------------------------------|---|-----|-------------|------|------|
| 1                                           | <pre>λ php artisan make:seeder LevelSeeder</pre> <p><b>INFO</b> Seeder [D:\laragon\www\WebLanjut\minggu3\PWLPOS\database\seeders\LevelSeeder.php] created successfully.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |               |            |            |            |            |                                             |   |     |               |      |      |                                             |   |     |         |      |      |                                             |   |     |             |      |      |
| 2                                           | <pre>&lt;?php  namespace Database\Seeders;  use Illuminate\Database\Console\Seeds\WithoutModelEvents; use Illuminate\Database\Seeder; use Illuminate\Database\Support\Facades\DB;  class LevelSeeder extends Seeder {     /**      * Run the database seeds.      */     public function run(): void     {         \$data = [             ['level_id' =&gt; 1, 'level_kode' =&gt; 'ADM', 'level_nama' =&gt; 'Administrator'],             ['level_id' =&gt; 2, 'level_kode' =&gt; 'MNG', 'level_nama' =&gt; 'Manager'],             ['level_id' =&gt; 3, 'level_kode' =&gt; 'STF', 'level_nama' =&gt; 'Staff/Kasir'],         ];         DB::table('m_level')-&gt;insert(\$data);     } }</pre> |            |               |            |            |            |            |                                             |   |     |               |      |      |                                             |   |     |         |      |      |                                             |   |     |             |      |      |
| 3                                           | <pre>D:\laragon\www\WebLanjut\minggu3\PWLPOS(main -&gt; origin) λ php artisan db:seed --class=LevelSeeder</pre> <p><b>INFO</b> Seeding database.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |               |            |            |            |            |                                             |   |     |               |      |      |                                             |   |     |         |      |      |                                             |   |     |             |      |      |
| 4                                           | <table><tr><th></th><th>level_id</th><th>level_kode</th><th>level_nama</th><th>created_at</th><th>updated_at</th></tr><tr><td><input type="checkbox"/>  Edit  Copy  Delete</td><td>1</td><td>ADM</td><td>Administrator</td><td>NULL</td><td>NULL</td></tr><tr><td><input type="checkbox"/>  Edit  Copy  Delete</td><td>2</td><td>MNG</td><td>Manager</td><td>NULL</td><td>NULL</td></tr><tr><td><input type="checkbox"/>  Edit  Copy  Delete</td><td>3</td><td>STF</td><td>Staff/Kasir</td><td>NULL</td><td>NULL</td></tr></table>                                                                                                                                                              |            | level_id      | level_kode | level_nama | created_at | updated_at | <input type="checkbox"/> Edit  Copy  Delete | 1 | ADM | Administrator | NULL | NULL | <input type="checkbox"/> Edit  Copy  Delete | 2 | MNG | Manager | NULL | NULL | <input type="checkbox"/> Edit  Copy  Delete | 3 | STF | Staff/Kasir | NULL | NULL |
|                                             | level_id                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | level_kode | level_nama    | created_at | updated_at |            |            |                                             |   |     |               |      |      |                                             |   |     |         |      |      |                                             |   |     |             |      |      |
| <input type="checkbox"/> Edit  Copy  Delete | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ADM        | Administrator | NULL       | NULL       |            |            |                                             |   |     |               |      |      |                                             |   |     |         |      |      |                                             |   |     |             |      |      |
| <input type="checkbox"/> Edit  Copy  Delete | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | MNG        | Manager       | NULL       | NULL       |            |            |                                             |   |     |               |      |      |                                             |   |     |         |      |      |                                             |   |     |             |      |      |
| <input type="checkbox"/> Edit  Copy  Delete | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | STF        | Staff/Kasir   | NULL       | NULL       |            |            |                                             |   |     |               |      |      |                                             |   |     |         |      |      |                                             |   |     |             |      |      |
| 5                                           | <pre>D:\laragon\www\WebLanjut\minggu3\PWLPOS(main -&gt; origin) λ php artisan make:seeder UserSeeder</pre> <p><b>INFO</b> Seeder [D:\laragon\www\WebLanjut\minggu3\PWLPOS\database\seeders\UserSeeder.php] created successfully.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |               |            |            |            |            |                                             |   |     |               |      |      |                                             |   |     |         |      |      |                                             |   |     |             |      |      |



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```
class UserSeeder extends Seeder
{
    /**
     * Run the database seeds.
     */
    public function run(): void
    {
        $data = [
            [
                'user_id' => 1,
                'level_id' => 1,
                'username' => 'admin',
                'nama' => 'Administrator',
                'password' => Hash::make('12345'),
            ],
            [
                'user_id' => 2,
                'level_id' => 2,
                'username' => 'manager',
                'nama' => 'Manager',
                'password' => Hash::make('12345'),
            ],
            [
                'user_id' => 3,
                'level_id' => 3,
                'username' => 'staff',
                'nama' => 'Staff/Kasir',
                'password' => Hash::make('12345'),
            ],
        ];
        DB::table('m_user')->insert($data);
    }
}
```

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|                          |                  | user_id | level_id | username | nama          | password                                                 | created_at | updated_at |
|--------------------------|------------------|---------|----------|----------|---------------|----------------------------------------------------------|------------|------------|
| <input type="checkbox"/> | Edit Copy Delete | 1       | 1        | admin    | Administrator | \$2y\$12\$oeWk87UdDKJtkYUhc/6ZTOgMJMVPDCI02/w7rv/JM9...  | NULL       | NULL       |
| <input type="checkbox"/> | Edit Copy Delete | 2       | 2        | manager  | Manager       | \$2y\$12\$2VNQLspiTnckq5T8uzfdvO0bHouz5z52kd1Jjhrvp3Z... | NULL       | NULL       |
| <input type="checkbox"/> | Edit Copy Delete | 3       | 3        | staff    | Staff/Kasir   | \$2y\$12\$lq0RzI0oXcTrDDEo94cCjunqgE/U2ccPu.y5b55cJZL... | NULL       | NULL       |

9

10

| Table                                           | Action                                      | Rows |
|-------------------------------------------------|---------------------------------------------|------|
| <input type="checkbox"/> failed_jobs            | ★ Browse Structure Search Insert Empty Drop | 0    |
| <input type="checkbox"/> migrations             | ★ Browse Structure Search Insert Empty Drop | 16   |
| <input type="checkbox"/> m_barang               | ★ Browse Structure Search Insert Empty Drop | 10   |
| <input type="checkbox"/> m_kategori             | ★ Browse Structure Search Insert Empty Drop | 5    |
| <input type="checkbox"/> m_level                | ★ Browse Structure Search Insert Empty Drop | 3    |
| <input type="checkbox"/> m_user                 | ★ Browse Structure Search Insert Empty Drop | 3    |
| <input type="checkbox"/> password_reset_tokens  | ★ Browse Structure Search Insert Empty Drop | 0    |
| <input type="checkbox"/> personal_access_tokens | ★ Browse Structure Search Insert Empty Drop | 0    |
| <input type="checkbox"/> t_penjualan            | ★ Browse Structure Search Insert Empty Drop | 10   |
| <input type="checkbox"/> t_penjualan_detail     | ★ Browse Structure Search Insert Empty Drop | 30   |
| <input type="checkbox"/> t_stok                 | ★ Browse Structure Search Insert Empty Drop | 10   |
| <input type="checkbox"/> users                  | ★ Browse Structure Search Insert Empty Drop | 0    |



## Praktikum Ke-4 Facades DB

| Langkah                                                                                     | Jawaban/Deskripsi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |               |                     |            |            |            |                                                                                             |   |     |               |      |      |                                                                                             |   |     |         |      |      |                                                                                             |   |     |             |      |      |                                                                                             |   |     |           |                     |      |
|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------|---------------------|------------|------------|------------|---------------------------------------------------------------------------------------------|---|-----|---------------|------|------|---------------------------------------------------------------------------------------------|---|-----|---------|------|------|---------------------------------------------------------------------------------------------|---|-----|-------------|------|------|---------------------------------------------------------------------------------------------|---|-----|-----------|---------------------|------|
| 1                                                                                           | <pre>PS D:\laragon\www\WebLanjut\minggu3\PWLPOS&gt; php artisan make:controller LevelController</pre> <p><b>INFO</b> Controller [D:\laragon\www\WebLanjut\minggu3\PWLPOS\app\Http\Controllers\LevelController.php] created successfully.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |               |                     |            |            |            |                                                                                             |   |     |               |      |      |                                                                                             |   |     |         |      |      |                                                                                             |   |     |             |      |      |                                                                                             |   |     |           |                     |      |
| 2                                                                                           | <pre>Route::get('/level', [LevelController::class, 'index']);</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |               |                     |            |            |            |                                                                                             |   |     |               |      |      |                                                                                             |   |     |         |      |      |                                                                                             |   |     |             |      |      |                                                                                             |   |     |           |                     |      |
| 3                                                                                           | <pre>&lt;?php  namespace App\Http\Controllers; use Illuminate\Support\Facades\DB;  use Illuminate\Http\Request;  class LevelController extends Controller {     public function index()     {         DB::insert('insert into m_level(level_kode, level_nama, created_at) values(?, ?, ?)');         return "Inserts data baru berhasil";     } }</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |               |                     |            |            |            |                                                                                             |   |     |               |      |      |                                                                                             |   |     |         |      |      |                                                                                             |   |     |             |      |      |                                                                                             |   |     |           |                     |      |
| 4                                                                                           | <div>localhost/WebLanjut/minggu3/PWLPOS/public/level</div> <p>Inserts data baru berhasil</p> <table><thead><tr><th></th><th>level_id</th><th>level_kode</th><th>level_nama</th><th>created_at</th><th>updated_at</th></tr></thead><tbody><tr><td><input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete</td><td>1</td><td>ADM</td><td>Administrator</td><td>NULL</td><td>NULL</td></tr><tr><td><input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete</td><td>2</td><td>MNG</td><td>Manager</td><td>NULL</td><td>NULL</td></tr><tr><td><input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete</td><td>3</td><td>STF</td><td>Staff/Kasir</td><td>NULL</td><td>NULL</td></tr><tr><td><input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete</td><td>4</td><td>CUS</td><td>Pelanggan</td><td>2025-02-28 16:05:24</td><td>NULL</td></tr></tbody></table> |            | level_id      | level_kode          | level_nama | created_at | updated_at | <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 1 | ADM | Administrator | NULL | NULL | <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 2 | MNG | Manager | NULL | NULL | <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 3 | STF | Staff/Kasir | NULL | NULL | <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 4 | CUS | Pelanggan | 2025-02-28 16:05:24 | NULL |
|                                                                                             | level_id                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | level_kode | level_nama    | created_at          | updated_at |            |            |                                                                                             |   |     |               |      |      |                                                                                             |   |     |         |      |      |                                                                                             |   |     |             |      |      |                                                                                             |   |     |           |                     |      |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | ADM        | Administrator | NULL                | NULL       |            |            |                                                                                             |   |     |               |      |      |                                                                                             |   |     |         |      |      |                                                                                             |   |     |             |      |      |                                                                                             |   |     |           |                     |      |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | MNG        | Manager       | NULL                | NULL       |            |            |                                                                                             |   |     |               |      |      |                                                                                             |   |     |         |      |      |                                                                                             |   |     |             |      |      |                                                                                             |   |     |           |                     |      |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | STF        | Staff/Kasir   | NULL                | NULL       |            |            |                                                                                             |   |     |               |      |      |                                                                                             |   |     |         |      |      |                                                                                             |   |     |             |      |      |                                                                                             |   |     |           |                     |      |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | CUS        | Pelanggan     | 2025-02-28 16:05:24 | NULL       |            |            |                                                                                             |   |     |               |      |      |                                                                                             |   |     |         |      |      |                                                                                             |   |     |             |      |      |                                                                                             |   |     |           |                     |      |
| 5                                                                                           | <pre>&lt;?php  namespace App\Http\Controllers; use Illuminate\Support\Facades\DB;  use Illuminate\Http\Request;  class LevelController extends Controller {     public function index()     {         //DB::insert('insert into m_level(level_kode, level_nama, created_at) values(?, ?, ?)');         //return "Inserts data baru berhasil";          \$row = DB::update('update m_level set level_nama =? where level_kode =?', ['Customer',         return "Update data berhasil. Jumlah data yang diupdate: ". \$row.' baris';     } }</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |               |                     |            |            |            |                                                                                             |   |     |               |      |      |                                                                                             |   |     |         |      |      |                                                                                             |   |     |             |      |      |                                                                                             |   |     |           |                     |      |



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localhost/WebLanjut/minggu3/PWLPOS/public/level

Update data berhasil. Jumlah data yang diupdate: 1 baris

|                          |      |      |        | level_id | level_kode | level_nama    | created_at          | updated_at |
|--------------------------|------|------|--------|----------|------------|---------------|---------------------|------------|
| <input type="checkbox"/> | Edit | Copy | Delete | 1        | ADM        | Administrator | NULL                | NULL       |
| <input type="checkbox"/> | Edit | Copy | Delete | 2        | MNG        | Manager       | NULL                | NULL       |
| <input type="checkbox"/> | Edit | Copy | Delete | 3        | STF        | Staff/Kasir   | NULL                | NULL       |
| <input type="checkbox"/> | Edit | Copy | Delete | 4        | CUS        | Customer      | 2025-02-28 16:05:24 | NULL       |

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localhost/WebLanjut/minggu3/PWLPOS/public/level

Delete data berhasil. Jumlah data yang dihapus: 1 baris

|                          |      |      |        | level_id | level_kode | level_nama    | created_at | updated_at |
|--------------------------|------|------|--------|----------|------------|---------------|------------|------------|
| <input type="checkbox"/> | Edit | Copy | Delete | 1        | ADM        | Administrator | NULL       | NULL       |
| <input type="checkbox"/> | Edit | Copy | Delete | 2        | MNG        | Manager       | NULL       | NULL       |
| <input type="checkbox"/> | Edit | Copy | Delete | 3        | STF        | Staff/Kasir   | NULL       | NULL       |

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## Data Level Pengguna

| ID | Kode Level | Nama Level    |
|----|------------|---------------|
| 1  | ADM        | Administrator |
| 2  | MNG        | Manager       |
| 3  | STF        | Staff/Kasir   |



## Praktikum Ke-5 Query Builder

| Langkah                                                                               | Jawaban/Deskripsi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |                        |                     |               |            |                                                                                       |   |     |            |      |                                                                                       |   |     |         |      |                                                                                       |   |     |         |      |                                                                                       |   |     |         |      |                                                                                       |   |     |                        |      |                                                                                       |   |     |                      |                     |
|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------|---------------------|---------------|------------|---------------------------------------------------------------------------------------|---|-----|------------|------|---------------------------------------------------------------------------------------|---|-----|---------|------|---------------------------------------------------------------------------------------|---|-----|---------|------|---------------------------------------------------------------------------------------|---|-----|---------|------|---------------------------------------------------------------------------------------|---|-----|------------------------|------|---------------------------------------------------------------------------------------|---|-----|----------------------|---------------------|
| 1                                                                                     | <pre>PS D:\laragon\www\WebLanjut\minggu3\PWLPOS&gt; php artisan make:controller KategoriController</pre> <p><b>INFO</b> Controller [D:\laragon\www\WebLanjut\minggu3\PWLPOS\app\Http\Controllers\KategoriController.php] created successfully.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |                        |                     |               |            |                                                                                       |   |     |            |      |                                                                                       |   |     |         |      |                                                                                       |   |     |         |      |                                                                                       |   |     |         |      |                                                                                       |   |     |                        |      |                                                                                       |   |     |                      |                     |
| 2                                                                                     | <pre>1 &lt;?php 2 3 use Illuminate\Support\Facades\Route; 4 use App\Http\Controllers\LevelController; 5 use App\Http\Controllers\KategoriController; 6 7 Route::get('/', function () { 8     return view('welcome'); 9 }); 10 11 Route::get('/level', [LevelController::class, 'index']); 12 Route::get('/kategori', [KategoriController::class, 'index']); 13</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |                        |                     |               |            |                                                                                       |   |     |            |      |                                                                                       |   |     |         |      |                                                                                       |   |     |         |      |                                                                                       |   |     |         |      |                                                                                       |   |     |                        |      |                                                                                       |   |     |                      |                     |
| 3                                                                                     | <pre>&lt;?php namespace App\Http\Controllers; use Illuminate\Support\Facades\DB; use Illuminate\Http\Request;  class KategoriController extends Controller {     public function index()     {         \$data = [             'kategori_kode' =&gt; 'SNK',             'kategori_nama' =&gt; 'Snack/Makanan Ringan',             'created_at' =&gt; now()         ];         DB::table('m_kategori')-&gt;insert(\$data);         return 'Insert data baru berhasil';     } }</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |                        |                     |               |            |                                                                                       |   |     |            |      |                                                                                       |   |     |         |      |                                                                                       |   |     |         |      |                                                                                       |   |     |         |      |                                                                                       |   |     |                        |      |                                                                                       |   |     |                      |                     |
| 4                                                                                     | <div>localhost/WebLanjut/minggu3/PWLPOS/public/kategori</div> <p>Insert data baru berhasil</p> <table><tr><th></th><th>kategori_id</th><th>kategori_kode</th><th>kategori_nama</th><th>created_at</th></tr><tr><td><input type="checkbox"/> Edit <input type="copy"/> Copy <input type="delete"/> Delete</td><td>1</td><td>ELK</td><td>Elektronik</td><td>NULL</td></tr><tr><td><input type="checkbox"/> Edit <input type="copy"/> Copy <input type="delete"/> Delete</td><td>2</td><td>PKN</td><td>Pakaian</td><td>NULL</td></tr><tr><td><input type="checkbox"/> Edit <input type="copy"/> Copy <input type="delete"/> Delete</td><td>3</td><td>MKN</td><td>Makanan</td><td>NULL</td></tr><tr><td><input type="checkbox"/> Edit <input type="copy"/> Copy <input type="delete"/> Delete</td><td>4</td><td>MNM</td><td>Minuman</td><td>NULL</td></tr><tr><td><input type="checkbox"/> Edit <input type="copy"/> Copy <input type="delete"/> Delete</td><td>5</td><td>PRT</td><td>Peralatan Rumah Tangga</td><td>NULL</td></tr><tr><td><input type="checkbox"/> Edit <input type="copy"/> Copy <input type="delete"/> Delete</td><td>6</td><td>SNK</td><td>Snack/Makanan Ringan</td><td>2025-02-28 16:58:27</td></tr></table> |               | kategori_id            | kategori_kode       | kategori_nama | created_at | <input type="checkbox"/> Edit <input type="copy"/> Copy <input type="delete"/> Delete | 1 | ELK | Elektronik | NULL | <input type="checkbox"/> Edit <input type="copy"/> Copy <input type="delete"/> Delete | 2 | PKN | Pakaian | NULL | <input type="checkbox"/> Edit <input type="copy"/> Copy <input type="delete"/> Delete | 3 | MKN | Makanan | NULL | <input type="checkbox"/> Edit <input type="copy"/> Copy <input type="delete"/> Delete | 4 | MNM | Minuman | NULL | <input type="checkbox"/> Edit <input type="copy"/> Copy <input type="delete"/> Delete | 5 | PRT | Peralatan Rumah Tangga | NULL | <input type="checkbox"/> Edit <input type="copy"/> Copy <input type="delete"/> Delete | 6 | SNK | Snack/Makanan Ringan | 2025-02-28 16:58:27 |
|                                                                                       | kategori_id                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | kategori_kode | kategori_nama          | created_at          |               |            |                                                                                       |   |     |            |      |                                                                                       |   |     |         |      |                                                                                       |   |     |         |      |                                                                                       |   |     |         |      |                                                                                       |   |     |                        |      |                                                                                       |   |     |                      |                     |
| <input type="checkbox"/> Edit <input type="copy"/> Copy <input type="delete"/> Delete | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ELK           | Elektronik             | NULL                |               |            |                                                                                       |   |     |            |      |                                                                                       |   |     |         |      |                                                                                       |   |     |         |      |                                                                                       |   |     |         |      |                                                                                       |   |     |                        |      |                                                                                       |   |     |                      |                     |
| <input type="checkbox"/> Edit <input type="copy"/> Copy <input type="delete"/> Delete | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | PKN           | Pakaian                | NULL                |               |            |                                                                                       |   |     |            |      |                                                                                       |   |     |         |      |                                                                                       |   |     |         |      |                                                                                       |   |     |         |      |                                                                                       |   |     |                        |      |                                                                                       |   |     |                      |                     |
| <input type="checkbox"/> Edit <input type="copy"/> Copy <input type="delete"/> Delete | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | MKN           | Makanan                | NULL                |               |            |                                                                                       |   |     |            |      |                                                                                       |   |     |         |      |                                                                                       |   |     |         |      |                                                                                       |   |     |         |      |                                                                                       |   |     |                        |      |                                                                                       |   |     |                      |                     |
| <input type="checkbox"/> Edit <input type="copy"/> Copy <input type="delete"/> Delete | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | MNM           | Minuman                | NULL                |               |            |                                                                                       |   |     |            |      |                                                                                       |   |     |         |      |                                                                                       |   |     |         |      |                                                                                       |   |     |         |      |                                                                                       |   |     |                        |      |                                                                                       |   |     |                      |                     |
| <input type="checkbox"/> Edit <input type="copy"/> Copy <input type="delete"/> Delete | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | PRT           | Peralatan Rumah Tangga | NULL                |               |            |                                                                                       |   |     |            |      |                                                                                       |   |     |         |      |                                                                                       |   |     |         |      |                                                                                       |   |     |         |      |                                                                                       |   |     |                        |      |                                                                                       |   |     |                      |                     |
| <input type="checkbox"/> Edit <input type="copy"/> Copy <input type="delete"/> Delete | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | SNK           | Snack/Makanan Ringan   | 2025-02-28 16:58:27 |               |            |                                                                                       |   |     |            |      |                                                                                       |   |     |         |      |                                                                                       |   |     |         |      |                                                                                       |   |     |         |      |                                                                                       |   |     |                        |      |                                                                                       |   |     |                      |                     |





5

```
<?php
namespace App\Http\Controllers;
use Illuminate\Support\Facades\DB;
use Illuminate\Http\Request;

class KategoriController extends Controller
{
    public function index()
    {
        $data = [
            'kategori_kode' => 'SNK',
            'kategori_nama' => 'Snack/Makanan Ringan',
            'created_at' => now()
        ];
        //DB::table('m_kategori')->insert($data);
        //return 'Insert data baru berhasil';

        $row = DB::table('m_kategori')->where('kategori_kode', 'SNK')->update(['kategori_nama' => 'Snack/Makanan Ringan']);
        return 'Update data berhasil. Jumlah data yang diupdate: ' . $row . ' baris';
    }
}
```

6

localhost/WebLanjut/minggu3/PWLPOS/public/kategori

Update data berhasil. Jumlah data yang diupdate: 1 baris

|                          |      |      | kategori_id | kategori_kode | kategori_nama          | created_at          |
|--------------------------|------|------|-------------|---------------|------------------------|---------------------|
| <input type="checkbox"/> | Edit | Copy | Delete      | 1 ELK         | Elektronik             | NULL                |
| <input type="checkbox"/> | Edit | Copy | Delete      | 2 PKN         | Pakaian                | NULL                |
| <input type="checkbox"/> | Edit | Copy | Delete      | 3 MKN         | Makanan                | NULL                |
| <input type="checkbox"/> | Edit | Copy | Delete      | 4 MNM         | Minuman                | NULL                |
| <input type="checkbox"/> | Edit | Copy | Delete      | 5 PRT         | Peralatan Rumah Tangga | NULL                |
| <input type="checkbox"/> | Edit | Copy | Delete      | 6 SNK         | Camilan                | 2025-02-28 16:58:27 |

7

localhost/WebLanjut/minggu3/PWLPOS/public/kategori

Update data dihapus. Jumlah data yang dihapus: 1 baris

|                          |      |      |        | kategori_id | kategori_kode          | kategori_nama | created_at |
|--------------------------|------|------|--------|-------------|------------------------|---------------|------------|
| <input type="checkbox"/> | Edit | Copy | Delete | 1 ELK       | Elektronik             | NULL          |            |
| <input type="checkbox"/> | Edit | Copy | Delete | 2 PKN       | Pakaian                | NULL          |            |
| <input type="checkbox"/> | Edit | Copy | Delete | 3 MKN       | Makanan                | NULL          |            |
| <input type="checkbox"/> | Edit | Copy | Delete | 4 MNM       | Minuman                | NULL          |            |
| <input type="checkbox"/> | Edit | Copy | Delete | 5 PRT       | Peralatan Rumah Tangga | NULL          |            |

9

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Data Kategori Barang</title>
</head>
<body>
    <h1>Data Kategori Barang</h1>
    <table border="1" cellpadding="2" cellspacing="0">
        <tr>
            <th>ID</th>
            <th>Kode Kategori</th>
            <th>Nama Kategori</th>
        </tr>
        @foreach ($data as $d)
            <tr>
                <td>{{ $d->kategori_id }}</td>
                <td>{{ $d->kategori_kode }}</td>
                <td>{{ $d->kategori_nama }}</td>
            </tr>
        @endforeach
    </table>
</body>
</html>
```



KEMENTERIAN PENDIDIKAN, KEBUDAYAAN, RISET, DAN TEKNOLOGI  
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**JURUSAN TEKNOLOGI INFORMASI**  
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<https://jti.polinema.ac.id>

10

## Data Kategori Barang

| ID | Kode Kategori | Nama Kategori          |
|----|---------------|------------------------|
| 1  | ELK           | Elektronik             |
| 2  | PKN           | Pakaian                |
| 3  | MKN           | Makanan                |
| 4  | MNM           | Minuman                |
| 5  | PRT           | Peralatan Rumah Tangga |



## Praktikum Ke-6 Eloquent ORM

| Langkah | Jawaban/Deskripsi                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | <pre>PS D:\laragon\www\WebLanjut\minggu3\PHLP05&gt; php artisan make:model UserModel</pre> <p>INFO Model [D:\laragon\www\WebLanjut\minggu3\PHLP05\app\Models\UserModel.php] created successfully</p>                                                                                                                                                                                                                                     |
| 3       | <pre>&lt;?php namespace App\Models;  use Illuminate\Database\Eloquent\Factories\HasFactory; use Illuminate\Database\Eloquent\Model;  class UserModel extends Model {     use HasFactory;      protected \$table = 'm_user';     protected \$primaryKey = 'user_id';</pre>                                                                                                                                                                |
| 4       | <pre>&lt;?php use Illuminate\Support\Facades\Route; use App\Http\Controllers\LevelController; use App\Http\Controllers\KategoriController; use App\Http\Controllers\UserController;  Route::get('/', function () {     return view('welcome'); });  Route::get('/level', [LevelController::class, 'index']); Route::get('/kategori', [KategoriController::class, 'index']); Route::get('/user', [UserController::class, 'index']);</pre> |
| 5       | <pre>&lt;?php namespace App\Http\Controllers;  use Illuminate\Http\Request; use App\Models\UserModels;  class UserController extends Controller {     public function index()     {         //coba akses model UserModel         \$user = UserModel::all(); // ambil semua data dari tabel m_u         return view('user', ['data' =&gt; \$user]);     } }</pre>                                                                         |



6

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Data User</title>
</head>
<body>
  <h1>Data User</h1>
  <table border="1" cellpadding="2" cellspacing="0">
    <tr>
      <th>ID</th>
      <th>Username</th>
      <th>Nama</th>
      <th>ID Level Pengguna</th>
    </tr>
    @foreach ($data as $d)
      <tr>
        <td>{{ $d->user_id}}</td>
        <td>{{ $d->username}}</td>
        <td>{{ $d->nama}}</td>
        <td>{{ $d->level_id}}</td>
      </tr>
    @endforeach
  </table>
</body>
</html>
```

7

## Data User

| ID | Username | Nama          | ID Level Pengguna |
|----|----------|---------------|-------------------|
| 1  | admin    | Administrator | 1                 |
| 2  | manager  | Manager       | 2                 |
| 3  | staff    | Staff/Kasir   | 3                 |

9

## Data User

| ID | Username   | Nama          | ID Level Pengguna |
|----|------------|---------------|-------------------|
| 1  | admin      | Administrator | 1                 |
| 2  | manager    | Manager       | 2                 |
| 3  | staff      | Staff/Kasir   | 3                 |
| 6  | customer-1 | Pelanggan     | 4                 |

11

## Data User

| ID | Username   | Nama              | ID Level Pengguna |
|----|------------|-------------------|-------------------|
| 1  | admin      | Administrator     | 1                 |
| 2  | manager    | Manager           | 2                 |
| 3  | staff      | Staff/Kasir       | 3                 |
| 6  | customer-1 | Pelanggan Pertama | 4                 |



## PERTANYAAN

Jawablah pertanyaan berikut sesuai pemahaman materi di atas

1. Pada Praktikum 1 - Tahap 5, apakah fungsi dari APP\_KEY pada file setting .env Laravel?

**Jawab:** APP\_KEY pada file .env berfungsi untuk **keamanan dalam enkripsi data**, termasuk hashing password, token autentikasi, dan session cookies. Laravel menggunakan APP\_KEY untuk mengenkripsi data sensitif agar tidak dapat dibaca oleh pihak yang tidak berwenang. Misalnya, jika aplikasi menggunakan fitur **enkripsi data** seperti encrypt() dan decrypt(), maka nilai APP\_KEY ini akan digunakan sebagai kunci untuk mengenkripsi dan mendekripsi data tersebut. Jika APP\_KEY tidak diset atau berubah setelah aplikasi berjalan, maka data terenkripsi sebelumnya tidak dapat didekripsi lagi. Oleh karena itu, **APP\_KEY sebaiknya tidak diubah setelah aplikasi diproduksi**.

2. Pada Praktikum 1, bagaimana kita men-generate nilai untuk APP\_KEY?

**Jawab:** Untuk meng-generate nilai baru untuk APP\_KEY, gunakan perintah berikut di terminal:

```
php artisan key:generate
```

Perintah ini akan secara otomatis menghasilkan **kunci enkripsi 32 karakter** dan menyimpannya ke dalam file .env, menggantikan APP\_KEY lama.

3. Pada Praktikum 2.1 - Tahap 1, secara default Laravel memiliki berapa file migrasi? dan untuk apa saja file migrasi tersebut?

**Jawab:** Saat pertama kali membuat proyek Laravel, terdapat **tiga file migrasi default**, yaitu:

- **create\_users\_table.php**

Digunakan untuk membuat tabel users yang menyimpan informasi pengguna aplikasi, seperti nama, email, dan password.

- **create\_password\_resets\_table.php**

Digunakan untuk membuat tabel password\_resets, yang menyimpan token reset password pengguna. Token ini dikirimkan ke email pengguna ketika mereka ingin mengatur ulang kata sandi.

- **create\_failed\_jobs\_table.php**

Digunakan untuk menyimpan informasi tentang **job yang gagal dieksekusi** dalam Laravel Queue System. Ini berguna untuk debugging atau menjalankan ulang job yang gagal.

4. Secara default, file migrasi terdapat kode \$table->timestamps();, apa tujuan/output dari fungsi tersebut?

**Jawab:** Kode \$table->timestamps(); secara otomatis akan **menambahkan dua kolom ke dalam tabel**, yaitu:

- **created\_at** → Menyimpan waktu kapan data pertama kali dibuat.
- **updated\_at** → Menyimpan waktu kapan data terakhir diperbarui.

Fitur ini berguna untuk **tracking perubahan data**, terutama jika aplikasi perlu mencatat kapan data dibuat atau diubah.



5. Pada File Migrasi, terdapat fungsi `$table->id()`; Tipe data apa yang dihasilkan dari fungsi tersebut?

**Jawab:** Kode `$table->id()`; akan membuat kolom dengan tipe data **BIGINT UNSIGNED AUTO\_INCREMENT** yang secara otomatis berfungsi sebagai **Primary Key**.

6. Apa bedanya hasil migrasi pada table `m_level`, antara menggunakan `$table->id()`; dengan menggunakan `$table->id('level_id')`; ?

Jawab:

- **`$table->id()`**; → Laravel akan secara otomatis membuat kolom dengan nama `id`, tipe **BIGINT UNSIGNED**, dan **auto-increment**.
- **`$table->id('level_id')`**; → Laravel akan membuat primary key dengan nama yang kita tentukan (`level_id`) tetapi tetap memiliki sifat **auto-increment**.

7. Pada migration, Fungsi `->unique()` digunakan untuk apa?

**Jawab:** Fungsi `->unique()` digunakan untuk **mencegah adanya data duplikat dalam suatu kolom**. Maka, jika ada dua barang dengan `barang_kode` yang sama, database akan menolak penyimpanan data baru dan memunculkan error.

8. Pada Praktikum 2.2 - Tahap 2, kenapa kolom `level_id` pada tabel `m_user` menggunakan `$table->unsignedBigInteger('level_id')`, sedangkan kolom `level_id` pada tabel `m_level` menggunakan `$table->id('level_id')` ?

**Jawab:**

- Pada tabel `m_level`, `level_id` adalah **primary key** sehingga menggunakan `$table->id('level_id')`.
- Pada tabel `m_user`, `level_id` berfungsi sebagai **foreign key**, sehingga harus menggunakan **`unsignedBigInteger`** agar **tipe datanya sama** dengan primary key di `m_level`.

9. Pada Praktikum 3 - Tahap 6, apa tujuan dari Class Hash? dan apa maksud dari kode program `Hash::make('1234')`;

**Jawab:** **Class Hash digunakan untuk meng-hash (mengacak) data sensitif**, seperti password, sehingga lebih aman. **Kode `Hash::make('1234')`**; akan menghasilkan password yang sudah di-hash menggunakan algoritma **bcrypt**.

10. Pada Praktikum 4 - Tahap 3/5/7, pada query builder terdapat tanda tanya (?), apa kegunaan dari tanda tanya (?) tersebut?

**Jawab:** Tanda tanya (?) dalam query digunakan untuk parameter binding. Dengan menggunakan tanda ?, nilai parameter akan diikat (bind) secara otomatis oleh Laravel sebelum dieksekusi ke database. Contohnya dalam query `DB::update('UPDATE m_level SET level_nama = ? WHERE level_kode = ?', ['Customer', 'CUS'])`; nilai 'Customer' akan menggantikan ? pertama, dan 'CUS' akan menggantikan ? kedua, sehingga query menjadi lebih aman dan terhindar dari manipulasi langsung oleh pengguna.

11. Pada Praktikum 6 - Tahap 3, apa tujuan penulisan kode `protected $table = 'm_user'`; dan `protected`



`$primaryKey = 'user_id';` ?

**Jawab:** `protected $table = 'm_user';` → Memberitahu Laravel bahwa model ini berhubungan dengan tabel `m_user`. `protected $primaryKey = 'user_id';` → Memberitahu Laravel bahwa **primary key** tabel ini bukan **id**, tetapi **user\_id**.

12. Menurut kalian, lebih mudah menggunakan mana dalam melakukan operasi CRUD ke database (DB Façade / Query Builder / Eloquent ORM) ? jelaskan!

**Jawab:** Eloquent ORM lebih mudah digunakan karena sintaksnya sederhana dan berbasis objek, memungkinkan interaksi dengan database tanpa menulis query SQL secara manual. Dibandingkan **Query Builder** dan **DB Façade**, Eloquent lebih intuitif, mendukung fitur seperti **relationship**, **mutator**, dan **soft delete**, sehingga pengelolaan data lebih efisien dan terstruktur.